

ALPHA NODUS · GRAVITY

Gravity & the Imaging Workflow

The complete step-by-step guide: from the first fax to the final payment, with every term explained

What is inside

Book 1. The full journey, in 15 steps, with the screenshot of a real order decoded, the integration plumbing, the market and regulation, a full A–Z glossary, and your team's working vocabulary (APAS, PASQ, auth terms, support terms).

Book 2. The document-to-order stage, screen by screen: the Documents list and filters, the review screen, the printed order summary, and the Orders list.

Prepared for Tushant Suneja · 8 October 2026

Built from the domain mastery guide (Gravity source material dated May 2026), live-product screenshots, and the team glossary.

How to read this guide

Every statement is tagged by how sure we are about it. Look for the coloured dots:

Dot	Meaning	How much to trust it
● Guide	From your domain mastery guide (built from Gravity's internal knowledge base, last modified 14 May 2026)	High. Confirm anything you plan to publish with engineering.
● Screen	Read directly from a screenshot or the printed order	High. This is what the live product showed.
● Team	From your team's glossary	High
● Inferred	General healthcare knowledge, or my best reading of how something works	Medium. Confirm with your team before repeating it externally.

How the books are laid out

- Each book is split into **Parts**, and each part into **Steps** or sections.
- Terms are defined **the first time they appear**, in bold, and again in the A–Z glossaries (Book 1 Part 6, Part 9; Book 2 Part 8).
- Open questions for your team are collected at the end of each book.
- All patient details have been left out. The screens come from a sample tenant; never paste real patient data into prompts, spreadsheets or ad tools.

If you only have 10 minutes

1. Book 1, **Part 1** (why Gravity exists, the four Centers, the three modes)
2. Book 1, **Part 3 overview diagram** (the whole journey on one page)
3. Book 1, **Part 7** (pronunciation, five traps, sentences to use in meetings)
4. Book 1, **Part 9** (your team's words: PASQ, APAS, auth number and the rest)

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Book 1: The Full Journey, from Referral to Payment

What this is: one patient's scan followed from the first fax to the final payment. Every step explains what happens, who does it, what Gravity does, and which words people use for it. The terms are defined where they first appear. A full A–Z glossary is in Part 6.

Sources:

- **Guide** = your domain mastery guide.
- **Screenshot** = the Service and Diagnosis card you shared.
- **General industry knowledge** = standard healthcare practice that the guide doesn't state. These items are marked, so you know which ones to confirm with the team.

Related: Book 2 zooms into the document-to-order stage (Documents list, filters, review screen, printed order, Orders list).

Freshness caveat (from the guide): Gravity-specific mechanics come from internal documentation last modified 14 May 2026. Confirm anything you plan to publish with engineering.

Part 1. The big picture (read this first)

1.1 The one-sentence idea

An imaging center (MRI, CT, mammography and so on) loses money because **the scanner sits idle about 35% of the time while patients wait 2–4 weeks**. The delay comes from the **front-office work before the scan**: reading faxes, phoning patients, checking insurance, getting insurer approval. **Gravity replaces that human front-office labor with AI workers**.

1.2 Key numbers to remember

Fact	Number
Real utilization of imaging centers (measured by Alpha Nodus's sensors in 2018–19)	60–65%
What they assumed (semiconductor fab efficiency)	95–99%
Hotels/airlines	~85%
Patient wait time at those same centers	2–4 weeks
Fixed costs of a center	~70% (rent, scanner lease, salaries)
Variable costs	~30%
Effect of going from 65% to 70% utilization	~+5% revenue vs ~+1.5% cost, so almost all of it is margin
Handwritten orders	~30% of orders
Manual prior auth cost (CAQH Index)	\$12.88 and 24 min
Electronic prior auth cost	\$5.38

Fact	Number
Physician/staff time on prior auth (AMA 2025)	13 hrs/week
Practices with a dedicated prior-auth employee	40%
Denials from missing/inaccurate data (Experian Health)	50%
Denials from incomplete/inaccurate registration	32%
CMS estimated savings from CMS-0057-F	~\$15B over 10 years

1.3 The mental model: Uber, not a software upgrade

Uber owns order-taking, routing and payment. The restaurant only cooks. **Gravity** owns document intake, scheduling, eligibility, prior authorization and payment prep. The imaging center only scans.

Alpha Nodus calls this category an **AOS (Agentic Operations System)**. Their line: "A RIS is a system of orders. An RCM is a system of claims. An AOS is a system of work."

1.4 What Gravity never does

- It never stores or reads images (that's PACS and the radiologists).
- It never replaces the RIS or the EMR.
- It never does post-service claims submission or collections (that's RCM territory; Gravity integrates with RCM systems).

1.5 The four "Centers" (the main sections of the Gravity app)

Center	Job	Where it sits in the journey
Document Center	Reads faxes and referrals and turns them into structured orders	Step 1–3
Booking Center	AI voice, SMS and chat scheduling (the voice agent is called Sandra)	Step 8–9
Revenue Center	Eligibility, benefits, prior auth, patient estimate (the prior-auth agent is Susan , inside GravityAuth)	Step 4–7
Flow Center	Recalls and referring-physician outreach	Step 14 and ongoing

1.6 The three operating modes

1. **Manual.** A human does everything. This is the starting state at about 95%+ of centers.
2. **Sidekick / assistive.** The AI does the work and a human reviews it before it ships. One scheduler goes from about 10 orders/hour to about 50/hour (5x). Almost every customer starts here.
3. **Fully agentic / autonomous.** The AI handles 80–95% of volume and humans handle only exceptions. A 50-person scheduling team becomes about 10 people, paid more, handling the hardest 5–20% of cases. The honest framing is "one person operating AI does the work of five."

Part 2. Decoding your screenshot (a real order card)

- This is a Gravity order page. The card is titled "**Service and Diagnosis.**" Here is every element.

2.1 The left block: what was ordered

Field	Value	Meaning
Service	77067 MA SCREENING BILATERAL TOMO W/CAD	77067 is the CPT code, the procedure code. "MA" is likely the center's modality/department abbreviation for mammography, but this is an assumption. It is a screening mammogram, both breasts, using tomosynthesis (3D mammography, "tomo") with CAD (Computer-Aided Detection software) .
Service Category	MG	Mammography, the imaging category. (Likely short for "mammogram", which is an inference.)
Diagnosis	Z12.31	The ICD-10 code, the "why." Z12.31 is the code for a routine screening mammogram, meaning no symptoms and routine cancer screening. ●
Location	River City Imaging Center Oak Sprawl	The Facility/Location where the scan will happen.
Date / Time	Oct 08, 2026, 8:00 am	The appointment Slot . It is the same as today, so this patient has just arrived.
Priority	Routine	Not STAT (urgent/immediate). Routine means normal scheduling.
RCI883325	italic ID	The Appointment ID (confirmed by the "Appointment ID" column in the Orders list). ● The "RCI" prefix probably identifies the center's RIS.

Why this combination matters. CPT (what) plus ICD-10 (why) is the pair the insurer uses to judge **medical necessity**. Screening mammograms are generally covered preventive care. ● That is why the **Estimate shows \$0** and **Authorized is skipped** in the timeline.

2.2 The top right

- **Study Status: Tech Start.** The scan itself is underway. The technologist has started the exam. A later screenshot of the same order shows **Completed**, then **Study Status: Unread** (images done, waiting for the radiologist to read them). Confirm the full status list with the team.
- **"..." menu.** Row-level actions (● assumption).
- The **tag icon** is a label/flag for the order (● assumption).

2.3 The timeline across the middle (the order's lifecycle)

Stage	In your screenshot	Meaning
Created	17 days ago, by a staff member	The Order record was made. (Either a human keyed it, or Auto Order Creation did.)
Booked	16 days ago, by System	An Appointment/Booking was set on a Resource (a room/scanner). "by System" means automation, likely Sandra or an integration. ● inferred
Estimate	\$0 , 3 days ago, by a staff member	The patient cost estimate (Good Faith Estimate). The patient owes \$0.
Authorized	(grey, empty)	Prior authorization. It is greyed out because it wasn't needed or wasn't done. A screening mammogram usually has no PA requirement. ●
Checked In	3 minutes ago, by System	The patient arrived and was checked in (● likely via kiosk or integration).
Completed ...	(cut off, grey)	The study is Complete , with more stages to the right that are cropped off.

The stages do not have to be strictly sequential (Estimate came *after* Booked, and Authorized was skipped).

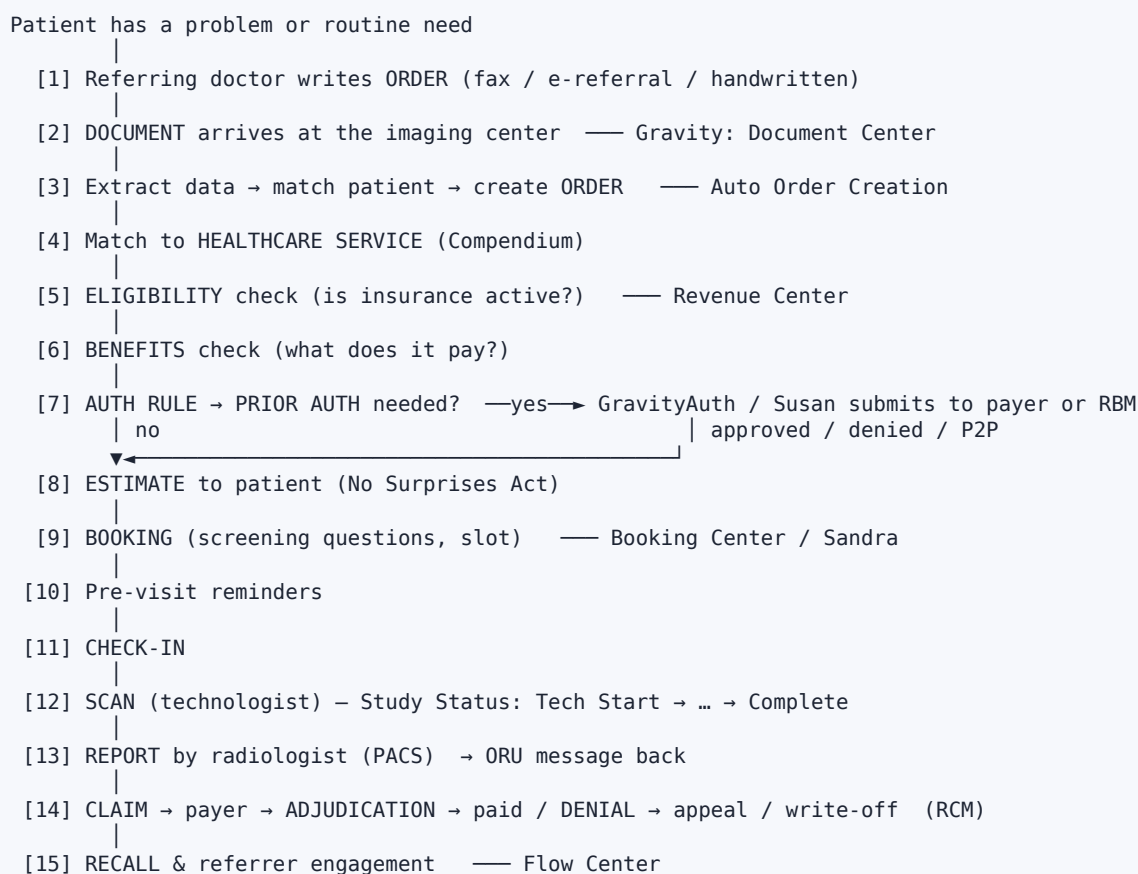
2.4 The tabs at the bottom (where each workflow lives)

Booking · Estimate · Prior Authorization · Documents · Check-In · Reports · Charges · Timeline

- **Booking:** appointment details and screening answers. On the sample order it shows a padlock: "Please contact Support to unlock this service", so the Booking module is not enabled for that tenant. See Book 2, Part 5.6.
- **Estimate:** the patient's cost calculation.
- **Prior Authorization:** the PA submission and result.
- **Documents:** the original referral/fax and attachments.
- **Check-In:** arrival and registration.
- **Reports:** the radiologist's result.
- **Charges:** what gets billed after service. •
- **Timeline:** the full audit log of every action.
- **Comment:** staff notes.

Part 3. The full journey: every step from start to finish

Overview diagram



STEP 0: Who is involved? (cast of characters)

Role	What they do	Selling note
Patient	Needs the scan	
PCP (primary care physician)	Writes most referrals	
Specialist	Cardiologist, orthopedist and others also refer	
Referring / ordering provider	The doctor who sent the patient. Imaging centers don't create their own demand, referring doctors do.	Key to Flow Center
Radiologist	Reads the images (usually remote, often at multiple centers)	Not your buyer
Technologist / tech / rad tech	Operates the scanner	
Sonographer	Ultrasound tech	
Scheduler	Books appointments	
PA specialist / VOB specialist	Gets prior auths and verifies insurance	
Practice administrator / manager	Typical buyer	
COO / Director of Operations	Strong internal champion	
RCM Director / Billing Manager	Owens denials, days in A/R, PA cycle time	
IT / IS Director	Integration, security and uptime gatekeeper	
Payer / payor	The insurance company (both spellings are used, so pick one per document)	
RBM	A company the insurer delegates imaging approvals to (eviCore, Carelon, and others)	Never a competitor

STEP 1: The referring doctor creates the order

What happens. The patient sees a doctor. The doctor decides imaging is needed and writes an **Order** (also called a **script** or **referral**). It contains:

- the patient's identity
- the exam wanted
- the diagnosis (ICD-10)
- priority
- insurance information
- clinical notes

How it arrives:

- **Fax** (still very common; HIPAA explicitly authorizes fax)
- e-referral from an **EMR/EHR**
- phone call
- handwritten script (~**30% of orders are handwritten**)

Terms

- **Order / script / referral:** the request for a clinical service. In Gravity, **Order is the single most important record**. Scheduling, clinical work and every financial step hang off it.
 - **Progress notes:** the referring doctor's documentation. They can run 20+ pages.
 - **Clinical notes / "clinical's":** progress notes used to prove medical necessity. Chasing these from doctors is one of the most painful parts of the job. Gravity automates the fax request.
 - **EMR / EHR:** the doctor's electronic chart system (Epic, Cerner, athenahealth, ModMed). Gravity is not one of these.
 - **PHI (Protected Health Information):** any patient-identifying health data. It must never go in an ad platform, pixel, spreadsheet or prompt.
-

STEP 2: The document hits the imaging center (Document Center)

The manual way. A staff member reads the fax, works out what it is, finds or creates the patient in the **RIS**, and types everything in. This takes hours a day.

The problem. Conventional **OCR** (Optical Character Recognition) returns raw text and fails on handwriting. The common objection is "we already have OCR." Gravity's answer: OCR gives raw text, not structured fields attached to the right patient.

What Gravity's Document Center does:

1. **Classifies** the incoming document (referral? clinical note? something else?).
2. **Extracts structured fields:**
 - patient name
 - DOB
 - referring physician
 - **CPT** (what procedure)
 - **ICD-10** (why)
 - payer
 - subscriber ID

It uses computer vision, OCR and an **LLM** (large language model) on both printed and handwritten content.

3. **Matches** the document to the right patient chart, or creates a new one.
4. **Learns** from every human correction.

Terms

- **RIS (Radiology Information System):** the day-to-day operating system of an imaging center. It covers scheduling, worklists, reporting and billing hooks. **Gravity sits on top of it and never replaces it.**
 - **Worklist:** a filtered list of orders a user works through. Filters saved in Gravity can be selected inside the GravityAuth extension.
 - **Snooze:** hide an order, account or document from a worklist until a chosen date.
-

STEP 3: Order creation (Auto Order Creation)

What happens. Auto Order Creation builds a live Order automatically, but only if it can confidently resolve **all three** of:

1. the **patient**
2. the **referring provider**
3. the **exam**

If any one is uncertain, it **routes to a human** instead of creating a wrong or partial order.

Related settings and modes

- **Auto Patient Creation:** a separate, pairable setting. If the patient doesn't exist and this is on, Gravity creates them. If it is off, the case goes to a human.
- **Draft order workflow:** the lower-risk starting mode. A human validates Gravity's patient, provider and service picks before an order is created. **It is the recommended first step before turning on Auto Order Creation.**
- **Activation:** converts a newly auto-created "shell" record into a usable one. Identifiers are confirmed and mappings applied.

The data chain behind an order (all Gravity "Record Types")

Patient → Referring provider → Order → Healthcare Service (from Compendium)
→ Coverage → Eligibility & Benefits → Auth Rule → Prior Authorization
→ Appointment/Booking (on a Resource) → Report/Result → Claim → Paid or Denial

Record Types (full list): Patient · Order · Booking/Appointment · Report/Result · Providers · Facilities · Location · Healthcare Services · Resources · Insurance · Patient Insurance Coverage · Estimates · Prior Authorization · Availability · Fee Schedule.

STEP 4: Match the exam to the center's catalogue

- **Healthcare Service:** Gravity's record for a bookable/orderable exam (e.g., "MRI Brain with and without contrast"). In your screenshot it is "77067 MA SCREENING BILATERAL TOMO W/CAD."
 - **Compendium:** the center's full catalogue of Healthcare Services. Auto Order Creation matches the incoming exam description against it. **Keeping it clean, with no near-duplicate names, is what makes matching work.**
 - **CPT (Current Procedural Terminology):** the procedure code, owned by the AMA.
 - 70553 = MRI brain with and without contrast
 - 74177 = CT abdomen/pelvis with contrast
 - 77067 = screening mammogram (your screenshot)
 - **HCPCS:** extends CPT to drugs, supplies and non-physician services.
 - **Modifier:** a two-character code that adjusts a CPT (e.g., -26 professional read vs. -TC technical component). **A wrong modifier means a denied claim.**
 - **ICD-10:** the diagnosis code (why the service was done).
 - **Protocol:** the specific scan sequence (e.g., "brain MRI with contrast").
 - **Modality:** the broad imaging type. See Part 6.
-

STEP 5: Verify eligibility (Revenue Center), "Is the insurance active?"

What happens. Before anything financial, the center checks that the patient's policy is active right now. Mechanically it is like a credit-card authorization check.

Terms

- **Eligibility / VOB (Verification of Benefits):** checking that insurance is active and what it covers. **Step one of everything financial.**
- **Coverage:** *this patient's* specific policy (subscriber ID, group, effective dates). It is **not** the same as **Insurance**, which is the payer/plan directory.
- **Subscriber:** the person who holds the policy (often but not always the patient, e.g., a child on a parent's plan).
- **Subscriber ID / Member ID:** the number on the insurance card.
- **Subscriber prefix:** the first three characters of the ID. It often encodes the plan variant and is used for auth-rule matching.
- **Group number / name:** the employer plan the coverage came from.
- **COB (Coordination of Benefits):** the rules for which insurer pays first when a patient has more than one. Gravity labels them **P** (primary), **S** (secondary), **T** (tertiary).
- **270/271:** the EDI pair for eligibility. **270 = the question** ("is this patient covered?"), **271 = the answer.**
- **EDI (Electronic Data Interchange):** the older standardized format for insurance transactions.
- **Clearinghouse:** a middleman that routes claims and eligibility requests between providers and hundreds of payers. **Availity** is the one Gravity uses.
- **EDI mapping:** the Availity payer ID that routes a request to a specific insurer. Without it, coverage shows **"Payor Not Activated"** and checks can't run.
- **Sync Lock:** a padlock on a coverage row. It means a subscriber ID was manually edited and is protected from being overwritten by an automated feed.
- **Payer mix:** the proportion of patients by insurance type. A heavy Medicare Advantage mix is a buying signal for prior-auth pain.
- **Medicare vs. Medicaid vs. Medicare Advantage (MA):**
 - **Medicare:** federal, mainly 65+.
 - **Medicare Advantage:** the privately run version. It prior-auths far more aggressively.
 - **Medicaid:** state-run, low-income, rules vary by state.

STEP 6: Benefits check, "What will it actually pay?"

- **Benefits:** what the plan actually pays.
 - **Copay:** a flat per-visit fee.
 - **Coinsurance:** a percentage the patient pays *after* the deductible.
 - **Deductible:** what the patient pays out of pocket before insurance starts paying. It is commonly 5,000–15,000 now.
 - **Out-of-pocket maximum:** the annual ceiling. After it is hit, insurance covers the rest of the year.
 - **In-network vs. out-of-network:** whether the center has a contract with that insurer. Out-of-network costs the patient far more.
 - **Allowed amount:** the contracted price the insurer recognizes (not the sticker price). Patient responsibility is calculated from it.
 - **Fee Schedule:** contracted prices per service, per payer. It feeds estimates.
-

STEP 7: Prior authorization (the headline feature)

7a. Does this exam need prior auth?

- **Prior Authorization (PA / prior auth / pre-auth / pre-cert):** the insurer's mandatory pre-approval before it will pay for a scan. Advanced imaging (MRI, CT, PET) is one of the most heavily prior-authed categories in US healthcare. **This is Gravity's wedge.** Say "prior auth," not "pre-cert."
- **Auth Rule:** a record saying whether PA is required for a **payer + group + plan-type + service** combination. Values: **Auth Required / No Auth Required / Unknown.**
- **Global rule vs. local rule:**
 - **Global rules:** pre-loaded and maintained by Gravity for common plans.
 - **Local rules:** created by the customer and take **precedence** over global. Customers can't edit global rules.
- **Wildcard:** a blank field in an auth rule that matches *any* value. A rule with only a payer name and service category applies across every group number and subscriber prefix.
- **Honest claim:** "pre-loaded rules cover common plans, and your team can add rules for anything specific to you." Do not claim "we know exactly when your patients need PA."

In your screenshot the **Authorized** stage is grey. The auth rule probably said no auth was required for this payer and screening mammogram. ● (confirm)

Team terms for this step: **PASQ** (check whether an auth already exists) and **APAS** (assisted submission) are explained in Part 9, with the inputs a submission needs (NPI, subscriber ID, ICD-10, CPT, payer).

7b. If yes, who approves?

- **RBM (Radiology Benefit Manager):** a company an insurer hires to decide imaging approvals. **Your customers submit PA requests TO them. They are never competitors.**
 - **eviCore** (Evernorth/Cigna, the biggest, say "EV-ee-core")
 - **Carelon Medical Benefits Management** (formerly AIM Specialty Health, under Elevance)
 - **NIA / RadMD** (Evolent)
 - **HealthHelp**
 - **Cohere Health** (AI-driven, has a supported GravityAuth portal, so closer to a partner surface)

7c. How Gravity submits it

- **GravityAuth:** a Chrome browser extension. Its AI agent **Susan** opens a real browser, logs into the payer's actual portal, finds the patient, fills in the form, submits and screenshots each step.
- **Susan:** the AI assistant inside GravityAuth. She can read the patient's progress notes and visit details and answer questions mid-workflow.
- **Why browser automation?** Most payer portals have no usable **API**, or charge per lookup. Automation clicks and types like a human. It identifies elements *semantically*, not by fixed screen position, so it tolerates portal redesigns.
- **API:** a structured way for two systems to talk directly in real time without a human clicking a screen.
- **278:** the EDI standard for **electronic** prior-auth submission. "Are you 278-enabled?" asks whether PA can be sent electronically instead of through a portal.

7d. What can come back

- **Authorization number:** issued when PA is approved. **It does not guarantee payment.**
- **Medical necessity:** the insurer's test for whether a scan is justified. A common denial reason is missing **conservative-care documentation** (payers usually want 4–6 weeks of physical therapy or other treatment documented before approving an MRI).

- **Denial → P2P (Peer-to-Peer):** the ordering physician must personally call the insurer's reviewing doctor and argue. It is expensive in physician time and disliked.
- **ABN (Advance Beneficiary Notice):** the Medicare form warning a patient they may have to pay, because Medicare probably won't cover the service.
- **ACR Appropriateness Criteria:** ACR's guidelines, used in medical-necessity decisions.

Cost of doing it manually: \$12.88 and 24 minutes each (CAQH), against \$5.38 electronic. Staff spend about 13 hours a week on PA (AMA).

STEP 8: Patient cost estimate

- **Estimate:** the pre-service cost calculation. Its output is **Patient responsibility** (what the patient owes).
- **Good Faith Estimate (GFE):** the pre-service estimate required under the **No Surprises Act (2022)**.
- **Surprise bill:** an unexpected patient balance, regulated by that Act.
- **How it's calculated:** Fee Schedule + Benefits (deductible remaining, copay, coinsurance, allowed amount).
- In your screenshot the **Estimate = \$0** (green), created 3 days ago.

STEP 9: Booking (Booking Center)

The problem. Most booking is still by phone, because MRI screening questions are a decision tree too complex for a static web form. The screening covers metal in the body, pacemaker, claustrophobia, contrast allergy, recent surgery and pregnancy.

What Gravity does. **Sandra**, an AI voice agent, answers 24/7. She authenticates the patient, runs the screening questions, matches an open slot, books it and confirms by text. Booking is also available over SMS and web chat. A human takes over with full context whenever the patient asks or the AI's confidence drops. Latency (the pause between turns) is an ongoing engineering focus.

Terms

- **Appointment / Booking:** the scheduled exam.
- **Slot:** a reservable time on a specific machine and tech.
- **Resource:** a capacity object (room, scanner, staff pool). It prevents capacity conflicts.
- **Availability:** open-slot data. **It is real-time via API only.** If a prospect's RIS can't expose availability over an API, that capability is harder to deliver well.
- **Contrast:** an injected imaging enhancer. It needs allergy and kidney screening.
- **No-show / Cancellation / Waitlist:** filling slots from the waitlist after cancellations is the highest-leverage and most badly done scheduling task.
- **Facility / Location:** the physical site.
- **POS (Place of Service):** a code for where care was delivered. It affects reimbursement.

STEP 10: Pre-visit (reminders and prep)

- General practice, not detailed in the guide. Reminders go out by text, call or email. The patient is told about prep (e.g., no deodorant for mammograms, fasting for some CTs) and what to bring. Flow Center's outreach engine (voice, text, email, fax, channel switching on no-response) covers this kind of contact.

STEP 11: Check-in

The patient arrives, identity and insurance are confirmed, and forms are signed. In your screenshot: **Checked In, 3 minutes ago, by System**. The **Check-In** tab holds these details. • The tab contents are not described in the guide.

STEP 12: The scan (clinical work, outside Gravity)

- The **technologist** performs the exam following the **protocol**.
- Your card shows **Study Status: Tech Start**.
- **Study**: a complete exam.
- **Series**: a sequence of images in a study (e.g., an MRI's T1/T2/FLAIR/diffusion).
- **Encounter**: a single visit.
- **DICOM**: the protocol scanners and PACS use to exchange images.
- **Gravity does not touch the images**.

Imaging modalities | Modality | What it is | |---|---| | X-ray / radiograph | 2D, cheapest | | CT | 3D X-ray, fast, good for bone/trauma | | MRI | Magnetic field + radio waves, high soft-tissue detail, almost always needs PA | | Ultrasound / sonography | Sound waves, no radiation, cheap | | DEXA | Bone density | | PET | Radioactive tracer, cancer staging | | PET/CT | Combined, very expensive | | Nuclear medicine | General radioactive-tracer category | | Fluoroscopy | Real-time X-ray video for guided procedures | | Mammography | Screening or diagnostic breast X-ray (your screenshot) | | Interventional radiology (IR) | Image-guided minimally invasive procedures (a specialty, not just a modality) |

STEP 13: Reading and reporting

- The images go to **PACS** (Picture Archiving and Communication System), where the **radiologist** reads them.
 - A **Report/Result** is produced and returns to the RIS and Gravity via an **ORU** message.
 - **TAT (Turnaround Time)**: scan completion to report delivery.
 - **MRN (Medical Record Number)**: the internal patient ID, unique within one organization only.
 - The **Reports** tab in your screenshot shows this.
-

STEP 14: Billing, claims and getting paid (RCM territory)

Gravity does the front end. The back end belongs to RCM systems, which Gravity integrates with.

- **RCM (Revenue Cycle Management)**: everything from scheduling through getting paid. Your buyer often has "revenue cycle" in their title.
- **Claim**: the bill sent to the insurer after service (**CMS-1500** professional, **UB-04** facility).
- **Adjudication**: the payer's processing of the claim (approve, partial or deny).
- **EOB (Explanation of Benefits)**: the payer's document explaining what was paid and why.
- **ERA (Electronic Remittance Advice)**: the electronic EOB, fed into the RCM system.
- **Denial**: the insurer refusing to pay. **The core problem Alpha Nodus sells against**.
- **Write-off**: revenue given up because a denial couldn't be overturned. Pure loss.
- **NPI (National Provider Identifier)**: a unique 10-digit ID. **Type 1 = individual, Type 2 = organization**.
- **Taxonomy code**: the classification of a provider's specialty, attached to their NPI.
- **KPIs**
 - **Days in A/R**: average days from billing to payment.

- **Denial rate:** % of claims denied on first pass.
- **Clean claim rate:** % accepted on first submission without rework.

Why denials trace back to the front desk. 50% of denials come from missing or inaccurate data, and 32% from incomplete or inaccurate registration (Experian Health survey). The usual causes are a wrong ICD-10/CPT pairing, a wrong subscriber ID or a wrong COB at intake. So fixing intake data (Document Center) reduces the biggest denial category.

The **Charges** tab in your screenshot is where this begins.

STEP 15: Recall and referrer engagement (Flow Center)

- **Recall:** a follow-up exam due at a set interval (e.g., a 6-month oncology surveillance MRI, or an annual screening mammogram like your screenshot's patient).
- Flow Center triggers outreach by voice, text, email and fax at the right interval. It escalates or switches channel on no-response, prioritizes whichever channel has worked for that patient, and lets the patient book straight from the message.
- **Referrer retention** is a key metric. Referring doctors drive demand.

Part 4. How the systems talk to each other (integration)

4.1 HL7 message types

Type	Stands for	Carries	Becomes in Gravity
ADT	Admit, Discharge, Transfer	Patient demographics, insurance	Patient, Coverage
ORM	Order message	Orders (optionally coverage)	Orders
SIU	Scheduling Information Unsolicited	Appointments	Bookings
ORU	Observation Result Unsolicited	Results	Reports/Results

- **HL7 (Health Level Seven):** the decades-old, pipe-delimited messaging standard (say "H-L-seven").
- **FHIR:** the modern REST/JSON successor (say "fire"). It is what **CMS-0057-F** mandates payers expose.
- **CDA:** a structured document format for clinical summaries.
- **Mirth:** an interface engine (middleware) that receives, transforms and routes HL7 messages. It is part of Alpha Nodus's stack.
- **Saying "we do HL7 and FHIR"** means we speak the old language every legacy RIS uses, and the new one payers are being forced onto.

4.2 Sync modes (defined per record type, not per integration)

Always ask, "which record type?"

Mode	Name	Rule	Risk
1	Read-only (one-way)	One writer, one listener	Lowest
2	Read + update	One system creates; the second can update but never create	Medium

Mode	Name	Rule	Risk
3	Full bi-directional	Both create and update	Highest; needs a stable ID or crosswalk, or you get duplicates

- **Stable unique ID:** permanent, never changes, and the only reliable cross-system match.
- **Fingerprint:** a match by name, DOB, phone or address. These can change or collide. Full bi-directional sync on fingerprints alone is **not advisable** (it "becomes an identity governance project").
- **ID crosswalk:** a stored mapping permanently linking a record in one system to its twin in another.
- **Seeding:** the one-time bulk load that populates a record type before ongoing sync.
- **Sync Profile:** the document describing, record type by record type, what each connected system may do, how records are matched and how data moves.
- **Practical rule:** never promise an implementation timeline (like "days, not months") without knowing whether the prospect's RIS has stable IDs.

4.3 Other systems in the ecosystem

- **PACS:** images and radiologist reading.
- **PM (Practice Management):** the RIS-equivalent for non-radiology specialties.
- **LIS (Laboratory Information System):** the RIS-equivalent for labs.
- **PIS (Pharmacy Information System):** the RIS-equivalent for pharmacy.
- **HIS (Hospital Information System):** a catch-all for hospital-wide IT.
- **RIS vendors:** **RamSoft** (PowerServer/OmegaAI, 750+ sites, already integrated, **the biggest untapped channel**), **ADS** (MedicsRIS, integrated, powers Bright Light Imaging), **eRAD**, **Merge** and **Epic Radiant**.

Part 5. Market, competition and regulation

5.1 Why nobody else does this

- **RIS vendors** are scheduling-software companies without an AI-workforce story.
- **RCM vendors** focus on post-service claims, so pre-service work is a bolt-on.
- **PACS vendors** serve radiologist workflow, and the front office isn't their domain.
- **AI voice/document startups** lack imaging workflow knowledge and RIS integrations.

5.2 Competitors: internal use only

Group	Who
Incumbent imaging prior auth	Infinx (software + services)
Billing with native PA	ImagineSoftware
AI-native entrants	Honey Health, Flexbone, Coral, Nakod
Broader RCM automation	Waystar, MD Clarity
Outsourced RCM firms	e.g., Healthcare Administrative Partners (the alternative Gravity loses to most often)
Availity	Two roles: Gravity's eligibility clearinghouse and a competitor in broader RCM automation

Counter-message for outsourcing: outsourcing moves the cost to someone else's staff, while automation removes it.

Register rule. To a prospect, say: "zero companies attack this from the patient-journey angle." Internally, name competitors accurately.

5.3 Regulation and forces

Item	What it is	Why it matters
HIPAA (1996)	Federal law protecting PHI (one P, two As)	A "HIPPA" typo is the biggest tell of an outsider
OCR (Office for Civil Rights)	HHS agency enforcing HIPAA (also the abbreviation for Optical Character Recognition, so context decides)	
BAA	Business Associate Agreement, required before a vendor handles PHI	
SOC 2 Type II	Audited security certification (controls tested over a period). Alpha Nodus holds it.	Never mention it in outbound copy (route to Shamit)
CMS-0057-F	2024 Interoperability and Prior Authorization Final Rule. Payers must run four FHIR APIs (Patient Access, Provider Access, Payer-to-Payer, Prior Authorization) by 1 Jan 2027 . Ops requirements began 1 Jan 2026. Payers publish PA metrics from 31 Mar 2026.	Best "why now" hook; ~\$15B savings over 10 years
CMS	Centers for Medicare & Medicaid Services, the federal agency setting rules	
MPFS	Medicare Physician Fee Schedule, annual pay update, flat-to-down in real terms	Why margin compression is permanent
No Surprises Act (2022)	Limits surprise out-of-network billing; requires Good Faith Estimates	Reason Gravity Estimate exists
MQSA	Mammography Quality Standards Act, requires facility certification; the FDA database is a source for mammography prospect lists	
Stark Law (1989)	Bans physician self-referral for financial gain	
AKS	Anti-Kickback Statute, bans healthcare kickbacks, with criminal penalties	
ACA (2010)	Mandated coverage, banned pre-existing-condition denials, indirectly drove the deductible/PA explosion	
HITECH (2009)	Drove EMR adoption via Meaningful Use incentives	

Part 6. Master glossary (A–Z within category)

A. Gravity vocabulary

Term	Meaning
Activation	Turning a newly auto-created shell record into a usable one
AOS	Agentic Operations System, Alpha Nodus's category name ("a system of work"). Spell it out on first external use.
Auth Rule	Whether PA is required for a payer + group + plan + service

Term	Meaning
Auto Order Creation	Builds an Order from a document; needs patient, referring provider and exam resolved, else routes to a human
Auto Patient Creation	Separate setting that creates new patients automatically or routes to a human
Booking Center / Document Center / Flow Center / Revenue Center	The four sections of the app
Center	A major section of the app
Compendium	The customer's catalogue of Healthcare Services
Coverage	This patient's specific policy
Draft order workflow	Human validates picks before order creation
EDI mapping	Availability payer ID for routing; absent means "Payor Not Activated"
Fee Schedule	Contracted price per service per payer
Global / local rule	Pre-loaded vs customer-created; local wins
GravityAuth	Chrome extension that automates PA
Healthcare Service	A bookable/orderable exam record
Order	The core request-for-service record
Record Type	A structured information category Gravity runs on
Resource	A room, scanner or staff pool
Sandra	The AI voice booking agent
Seeding	One-time bulk load
Snooze	Hide an item from a worklist until a date
Susan	The AI assistant inside GravityAuth
Sync Lock	Protects a manually edited subscriber ID
Sync Profile	Per-record-type sync rules document
Tenant	One customer organization (multi-tenant platform; "tenant-scoped" is limited to one customer)
Wildcard	Blank auth-rule field matching any value
Worklist	A filtered list of orders to work through

B. Integration and data

HL7 · ADT / ORM / SIU / ORU · FHIR · CDA · DICOM · Mirth · API · Sync modes 1/2/3 · Stable ID vs fingerprint · ID crosswalk · MRN · 278 · 270/271 · EDI · Clearinghouse (all defined in Part 3 and Part 4).

C. Coding and billing

Term	Meaning
CPT	Procedure code (what was done)
ICD-10	Diagnosis code (why)
HCPCS	Extends CPT to drugs and supplies

Term	Meaning
Modifier	2-character CPT adjuster (-26, -TC)
NPI	10-digit provider ID; Type 1 individual, Type 2 organization
Taxonomy code	Provider specialty classification
POS	Place of Service code

D. Insurance and revenue cycle

Term	Meaning
Payer / Payor	The insurance company
Provider	A person (doctor) or an organization (center); context decides
Referring / ordering provider	The doctor who sent the patient
Subscriber / ID / prefix / group	Policyholder / card number / first 3 chars / employer plan
COB (P/S/T)	Which insurer pays first
Eligibility (VOB)	Is the insurance active and what does it cover
Benefits	What the plan pays
Copay / coinsurance / deductible / OOP max	Flat fee / % after deductible / pay-first amount / annual cap
In-network / out-of-network	Contracted or not
Allowed amount	Contracted recognized price
Patient responsibility	What the patient owes
Claim	Bill to the insurer (CMS-1500 / UB-04)
Denial	Refusal to pay
Adjudication	Payer's claim processing
EOB / ERA	Payment explanation (paper / electronic)
Write-off	Revenue lost after an unwinnable denial
Surprise bill / GFE	Unexpected balance / required pre-service estimate
RCM	Revenue Cycle Management
Prior Authorization	Insurer pre-approval
Authorization number	Issued on approval, no payment guarantee
Medical necessity	Is the scan justified
Clinical notes / clinicals	Documentation proving necessity
P2P	Doctor-to-doctor appeal call
ABN	Medicare "you may have to pay" form
Days in A/R, Denial rate, Clean claim rate	RCM KPIs
Payer mix	Patient insurance-type breakdown

E. Compliance

HIPAA · PHI · BAA · SOC 2 Type II · Stark Law · AKS · ACA · HITECH · No Surprises Act · CMS-0057-F · MPFS · MQSA · OCR · CMS · Medicare/Medicaid/MA: see Part 5.3.

F. Software and players

RIS · PACS · RCM · EMR/EHR · PM · LIS · PIS · HIS · RBM, plus the RIS vendors, RBMs and competitors in Parts 4 and 5.

G. Roles

PCP · Specialist · Referring/ordering physician · Radiologist · Technologist (tech, rad tech) · Sonographer · Scheduler · PA/VOB specialist · Practice administrator/manager · COO/Director of Operations · RCM Director/Billing Manager · IT/IS Director. See Step 0.

H. Workflow terms

Order/Script/Referral · Progress notes · Encounter · Study · Series · Modality · Slot · Protocol · Contrast · No-show · Cancellation · Waitlist · Recall · STAT (urgent) · Priority (Routine/STAT).

I. Business metrics

Term	Meaning
Utilization rate	Imaging 60–65% vs hotels/airlines ~85%, fabs 95–99%
TAT	Turnaround time
Referrer retention	Keeping referring doctors sending patients
Patient no-show rate	
CAC	Customer acquisition cost
ACV	Annual contract value
LTV	Lifetime value
NRR	Net revenue retention

J. Industry bodies

Body	What
RSNA	Biggest conference, Chicago, late November
RBMA	Radiology Business Management Association (operations/business, buyer-heavy)
ACR	American College of Radiology; publishes Appropriateness Criteria
AHRA	Association for Medical Imaging Management
HIMSS	Broader healthcare IT conference
CAQH	Nonprofit alliance; publishes the CAQH Index of PA costs
AMA	American Medical Association; owns CPT, publishes the admin-burden survey

Part 7. Speak like an insider

7.1 Pronunciation

Written	Say
FHIR	"fire"
PACS	"packs"
HIPAA	"HIP-ah"
eviCore	"EV-ee-core"
Availity	"a-VAIL-ity"
HL7	"H-L-seven"
CPT, NPI	Spell the letters

7.2 Five traps

1. **HIPAA** has one P and two As.
2. Say "**prior auth**", not "pre-cert."
3. **Never** say Gravity "replaces" the RIS or EMR. It sits on top.
4. **Radiologists are not the buyer.** Administrators and revenue-cycle directors are.
5. RBMs (eviCore, Carelon, NIA, HealthHelp, Cohere) are **never competitors**.

7.3 Sentences you can use in a meeting

- "Where does the order enter, by fax, e-referral or handwritten? And what's the RIS?"
- "Is this a Mode 1, 2 or 3 sync, and for which record type?"
- "Does their RIS have stable unique IDs, or are we matching on fingerprints?"
- "Do they auto-create orders, or are they on the draft order workflow?"
- "What's their payer mix? A heavy Medicare Advantage share means a worse PA queue."
- "Is the payer 278-enabled, or does it need GravityAuth through the portal?"
- "Is the denial from front-end data (ICD-10/CPT pairing, subscriber ID, COB) or from medical necessity?"
- "Do local auth rules override the global ones for this payer?"
- "Is eligibility showing 'Payor Not Activated', meaning the EDI mapping is missing?"
- "Which state is the study in, and has the Estimate been generated?"

7.4 Reading your screenshot aloud

"This is a routine screening mammogram, CPT 77067, diagnosis Z12.31. Order created 17 days ago, booked 16 days ago by the system, a \$0 estimate generated 3 days ago. No prior auth on the timeline, which fits for screening. The patient checked in 3 minutes ago and the study status is Tech Start, so the tech has begun the exam."

Part 8. Open questions for your team (things I could not confirm)

1. What does RCI883325 represent? Answered: it is the **Appointment ID**.

2. What is the **full list of Study Status values**? Known so far: Tech Start, then Unread.
3. Is the **"MA"** in "77067 MA" the department or modality code, and is "MG" the Service Category value for mammography?
4. Is "by System" on **Booked** and **Checked In** Sandra, a kiosk, or an HL7 feed?
5. What do the **cropped timeline stages** to the right of "Complete" show?
6. What does the **tag icon** and the **"..." menu** do on the card?
7. Are **screening mammograms** configured with "No Auth Required" in the auth rules by default?
8. What is in the **Check-In** and **Charges** tabs?
9. Which of the Part 3 mechanics (automation percentages, rule behavior, product names) are still current as of today?

Part 9. Your team's working vocabulary, linked to the process

This part adds the vocabulary your team uses day to day. Source: the team glossary you provided (● **Team**). Notes marked ● are my inference. Each term is tied to the step and the screen where it appears.

9.1 The prior-auth inputs: where each one lives

A prior-auth request is built from a handful of data points. Each one is a term from your glossary and each has a home in Gravity.

Term (team definition)	Where it comes from	Where you see it in Gravity	Process step
Referring Physician / Ordering Provider: the doctor requesting the exam	The referral or fax	Provider section on the review screen; Ordering Provider card on the printed order	Step 1, 3
NPI Number (National Provider Identifier): the facility has one and the referring physician has one	Provider record; facility record	Ordering Provider card (Type 1, the doctor); printed order header (Type 2, the facility)	Step 3, 7
Facility / Place of Service: the servicing provider (our tenant)	The imaging center	Header of the printed order; tenant switcher at the top	Step 7, 12
Tenant: our customers, the radiology centers, the facility	The customer organization	The "River City Imaging Centers - New" dropdown	All
Payer: the health insurance (BCBS, United Healthcare, Cigna, Aetna and so on)	Insurance card or fax	Coverage card; Select Insurance list	Step 5, 7
Subscriber ID: the member ID on the insurance card, an identification number issued by the insurer, like a government-issued ID number	Insurance card	The long ID under the payer name on the Coverage card	Step 5, 7
ICD-10 Code (International Classification of Diseases, 10th revision), the diagnosis code. Examples: R51.9 , M54.12	The doctor's order	Diagnosis line on the order (e.g., Z12.31 on your sample)	Step 4, 7
CPT Code (Current Procedural Terminology), the "service type": a 5-digit number for the imaging exam	The doctor's order	Service line on the order (e.g., 77067 on your sample)	Step 4, 7

Two examples from the glossary, decoded ●: **R51.9** is "headache, unspecified" and **M54.12** is "radiculopathy, cervical region" (a pinched nerve in the neck). Both are typical reasons a doctor orders an MRI, and payers check that the diagnosis supports the exam.

Why this table matters. If any one of these is missing or wrong, the authorization request fails or gets denied. • The guide says 50% of denials trace to missing or inaccurate data, and 32% to registration errors. When someone says "the auth bounced," ask: *which of these eight was wrong?*

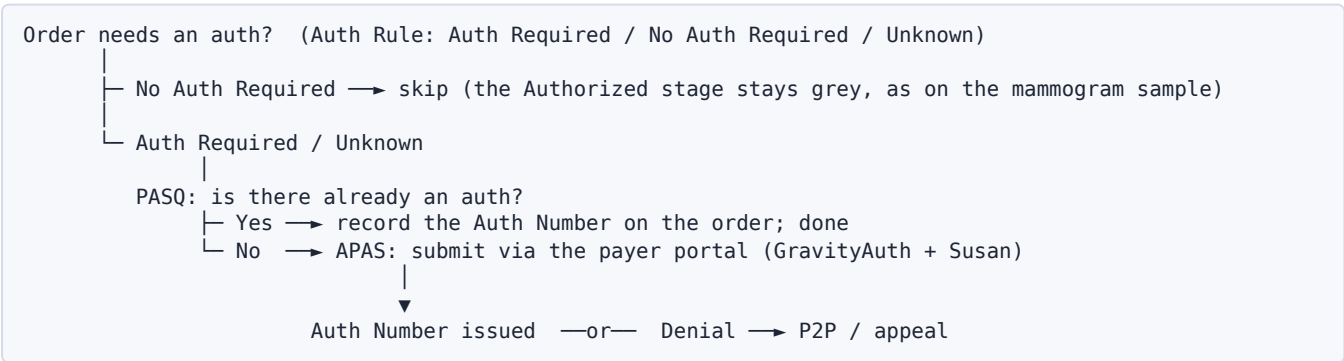
9.2 The authorization terms

Term	Team definition	Notes and links
Auth	Authorization, an approval from the insurance for an imaging exam	The same thing as prior authorization (Step 7). In Gravity it is the Authorized stage on the order timeline and the Prior Authorization tab.
Auth Number	Also called order ID # , reference # , tracking ID , case number . The number provided to identify and track an approval.	• Issued when a PA is approved. It does not guarantee payment . Different payers use different names for it, so expect all five labels.
Payer Portal	The insurance website where providers go to get authorizations	• This is where GravityAuth (the Chrome extension) and its AI assistant Susan log in, fill the form, submit and screenshot each step.
Gravity	Our application, which has our user interface and AI assistants	The user interface plus Susan (prior auth), Sandra (booking) and the Document Center AI.
UI (User Interface)	Our Gravity app	Everything you saw in your screenshots: Documents, Orders, review screen, tabs.

9.3 APAS and PASQ: the two prior-auth actions

Acronym	Stands for	Plain meaning	Where it fits
PASQ	Prior Auth Status Query	Check whether an authorization already exists (or what state it is in)	• Done first, before submitting, so the team does not file a duplicate. Also used later to see whether a pending request has been decided.
APAS	Assisted Prior Auth Submission	Submit a prior-auth request with Gravity assisting	• This is the Sidekick / assisted mode from the guide: the AI does the work (GravityAuth and Susan in the payer portal) and a person is involved. This is the "assistive" step between fully manual and fully autonomous.

Order of operations • (a logical reading of the definitions; confirm with your team):



9.4 Support and engineering vocabulary

Term	Team definition	How to use it
Release	When a stable version switches , for example from 3.10 to 3.11	"That fix ships in the next release." A release changes what customers see in the UI, so it is the point where behavior (and screenshots like yours) can change. • This is also why Gravity details need re-checking from time to time.
Critical Issue	A defect that affects productivity and interrupts the flow of operations	Use this word only when work is actually blocked (for example, auths cannot be submitted). It decides how urgent the response is.
Log a Bug	Use the support button in the Gravity app to report an issue affecting the auth process. Managers and the internal customer success team create a ticket in GitHub .	Two routes: (1) users click the support button in the UI, (2) managers or customer success open a GitHub ticket. The support button is the blue help icon at the bottom right of your screenshots •.

9.5 Putting it together: a sentence in the team's own language

"The **tenant** sent an **ordering provider's** referral. Gravity read the **CPT** and **ICD-10**, matched the **payer** and **subscriber ID**, and the auth rule says an **auth** is required. We ran a **PASQ** to see whether one already existed. It didn't, so we did an **APAS** through the **payer portal**. We got an **auth number** back. If the **UI** had blocked us, that would be a **critical issue**, and we would **log a bug** with the support button."

9.6 How these terms reconcile with earlier parts

- **Facility** here means the imaging center (our **tenant**). Do not confuse it with **POS (Place of Service)** in billing, which is a *code* for where care was given. Your team uses "Facility / Place of Service" for the servicing provider itself.
- **Auth Number** and **Authorization number** (Part 3, Step 7d) are the same thing.
- **Prior authorization** (guide term) = **Auth** (team shorthand). Say "prior auth" externally and "auth" internally.
- **Gravity** (team definition) = the **UI plus the AI assistants**, which matches the "four Centers" view in Part 1.

Built from your domain mastery guide, with your screenshot decoded in Part 2. • items are general industry knowledge, not from the guide.

Book 2: From Document to Order, Screen by Screen

Companion to: Book 1 (the full journey from referral to payment). This guide zooms into **the first third of that journey**, the part where a fax, photo or upload arrives and becomes an **Order**. It also decodes every screen you shared.

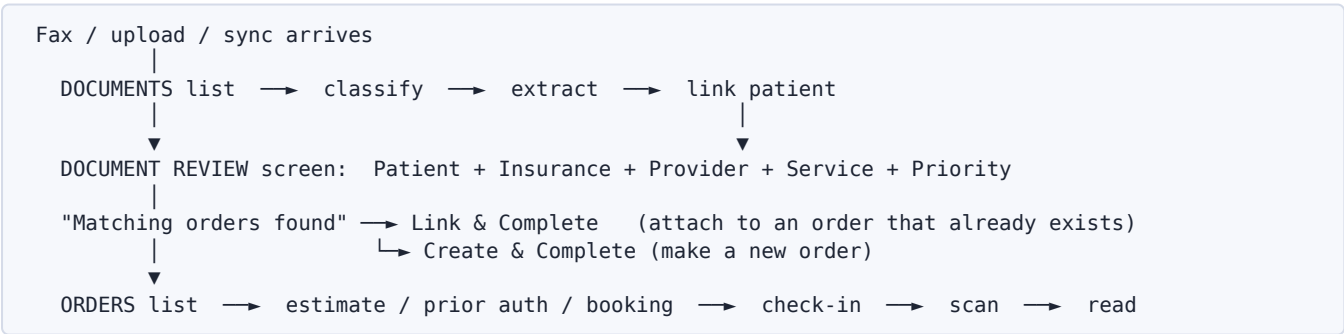
Source tags

- **Guide:** your domain mastery guide.
- **Screen:** read directly from your screenshots or the PDF printout.
- **Inferred:** my best reading of how it works. Confirm with your team before repeating it externally.

A note on the sample data. The screens come from the tenant "River City Imaging Centers - New", which looks like a test or demo tenant. I describe fields, not people. Never paste real patient details into a prompt, spreadsheet or ad platform (that is PHI, see the glossary).

Part 1. The three things you shared, and how they fit together

What you shared	What it is	Where it sits in the journey
Documents list (with filters)	The inbox. Every fax, photo and upload lands here.	Step 1–3 below
Document review screen (fax on the left, form on the right)	The workbench where a document is checked and turned into an order (or linked to one).	Step 4–10 below
Orders list and the printed order summary PDF	The result. An order that is scheduled, checked in, scanned and waiting to be read.	After Step 10



Part 2. The Documents list, field by field

2.1 Top navigation (on every screen) •

Item	Meaning
Documents	Inbox of incoming files (this guide's focus)
Orders	All orders (requests for exams)

Item	Meaning
Patients	Patient records
Bookings	Appointments (the Booking Center, Sandra's area)
Charges	The billing side after the service
Analytics	Reports and dashboards
"..."	More menu (hidden items on a narrow screen)
"River City Imaging Centers - New" dropdown	Tenant switcher. A tenant is one customer organization; Gravity is multi-tenant, so each customer's data is isolated. The URL also carries <code>?tenant=...</code> , which is why a link opens one customer's data only. •
Flask icon, bug icon	• Probably "labs / beta features" and "report an issue"
User name and green dot	Who is logged in (green means online)
"Powered by AlphaNodus"	The footer branding

2.2 Tabs and tools above the table •

Item	Meaning
All	Every document
Daily Practice (110) ★	A saved filter / worklist . The number is how many documents match right now. The star means it is a favourite. • <i>"Filters saved in Gravity become selectable inside GravityAuth."</i>
Search Documents	Free-text search (patient name and so on)
Filters	Opens the filter panel (see 2.4)
Last 90 Days	Date range for the list
+ Upload document	A person manually adds a file (this becomes the Web Upload source)
Refresh icon	Reloads the list
"1 new document" blue pill	New documents arrived since you loaded the page. Click to load them. Documents arrive continuously.
Displaying 1–20 of 26,616 records, 1331 pages, 20 / page	Total volume in the last 90 days for this tenant. This shows how big the inbox is.

2.3 The columns •

Column	What it shows	Why it matters
Checkbox	Select rows for bulk actions •	
Patient Name (underlined link)	The patient the document was matched to	If there is a chain-link icon beside the name, the document is linked to a patient chart. A name with no link icon is not yet linked.
Visit	A visit ID such as GV... (blank on unlinked rows)	Ties the document to an encounter •
Birth Date	DOB extracted from the document	Used to match the patient
Source	How the document got in	See the list of sources below

Column	What it shows	Why it matters
File Type	application/pdf or image/jpeg	PDFs are usually faxes; JPEGs are usually phone photos
Tags	Free labels on the document	Used for filtering
Category	What kind of document the AI thinks it is (blue "Order" pill on the first row)	Decides what happens next
Assignees	The staff member responsible	Empty means nobody owns it yet
Created	How long ago it arrived ("a minute ago")	Fax-to-order speed starts here
Action: Process button	A person opens the document to work it	The manual route
Action: orange sparkle button	The AI action (the sparkle icon is Gravity's AI marker)	Likely runs or re-runs AI processing on that document

The four JPEG rows in your screenshot are the same patient and the same visit ID, uploaded from a phone within the same minute. • This is most likely a patient or staff member photographing several pages (an insurance card front and back, ID and so on).

2.4 The Filters panel (every filter explained) •

Filter	Options seen	Meaning
Saved filters	"Select filter", Clear	Reuse or reset a stored combination of filters
Assignee	dropdown	Show only one person's work
Tags	dropdown	Filter by label
Source	Document Sync · Gravity User Portal · Inbound Fax · Mobile Upload · Prior Auth · System · Web Upload (each with Include / Exclude)	See the source table below
File type	dropdown	PDF, JPEG and so on
Category	None · Identification · Insurance Card · Lab Report · Letter Of Protection · Order · Prior Auth · Progress Note (list may continue)	What the document is
Category Status	Assigned · Auto Assigned · Human Review · In Progress (list may continue)	How the category was decided or where it is in handling
Status	"Select status"	Processing state of the document
Order created	Yes / No / All	Has this document already produced an order? Choose No to see work still waiting.
Patient Linking	Yes / No / All	Has the document been tied to a patient chart? Choose No to see documents the system could not match.
Hide Snoozed	Yes / No (Yes is on by default in your screenshot)	Hides documents someone has snoozed until a later date
Filter name / Cancel / Update Filter		Save this combination under a name

Include vs Exclude (toggle at the top of the Source, Category and Category Status lists): *Include* means "show only these". *Exclude* means "show everything except these".

2.5 The seven document sources • (meanings • unless noted)

Source	Plain meaning
Inbound Fax	A fax received by the center's fax number. Still the most common way referrals arrive. • HIPAA explicitly authorizes fax.
Mobile Upload	A photo taken on a phone and uploaded (insurance card, ID, paper script)
Web Upload	A person clicked Upload document in the web app
Gravity User Portal	Uploaded through a Gravity portal login • (possibly a referrer or patient portal)
Document Sync	Pulled in automatically from another system (RIS, EMR or a file drop). This is the integration route.
Prior Auth	A document created by the prior-authorization workflow (approval letters, payer replies, screenshots)
System	A document generated by Gravity itself

2.6 The eight document categories • (meanings • unless noted)

Category	What it is	What usually happens next
Order	A request for an exam (a script or referral)	Becomes an Order record. This is the main path of this guide.
Progress Note	The referring doctor's clinical notes. • It can run 20+ pages.	Attached to the order as proof of medical necessity (needed for prior auth)
Insurance Card	Photo of the card	Used to create or update the patient's Coverage
Identification	Driver's licence or other ID	Used to confirm patient identity
Prior Auth	A prior-authorization document	Attached to the authorization
Letter Of Protection	A contract used in injury or accident cases, where an attorney guarantees the provider will be paid from a settlement •	Changes who pays (not the insurer)
Lab Report	Lab results, sometimes needed to justify a scan	Attached to the order
None	Not classified	Goes to a human

2.7 Category Status values (my reading) •

Status	Likely meaning
Auto Assigned	The AI categorized the document on its own, confidently
Human Review	The AI was not confident, so a person must decide (this matches the guide's rule that low-confidence cases go to a human)
Assigned	The document has been given to a specific staff member
In Progress	Someone is working on it now

Part 3. The Document Review screen: where a document becomes an order

3.1 Layout •

Left side: the document viewer.

- Page thumbnails (the "1" badge means one page).
- Buttons for rotate, fit-to-width and fit-to-height, flip or rotate pages, zoom out, zoom in, and previous / next document.
- The fax itself, with the machine-printed header ("Oct/8/2026 7:45:26 AM" and the sender's fax label).

Right side: the work form.

- Top icons (left to right): comment, AI sparkle, download, copy, history, share, tag, assign to a person, timeline.
 - (Most of these are standard; confirm the exact functions.)
- **Five tabs: Order · Progress Note · ID · Insurance · More.** Each tab lets staff handle a different kind of information from the same document. The **Order** tab is the one in your screenshots.

3.2 "Matching orders found: 7" •

A blue badge says **7 matching orders were found**. Gravity searched for orders that might already match this document (same patient, similar exam or dates) so staff do not create duplicates.

Two buttons sit beside it, both greyed out until the form is complete: | Button | What it does | |---|---| | **Link & Complete** | Attach this document to an order that **already exists**, then mark the document finished | | **Create & Complete** | Create a **new order** from this document, then mark the document finished |

Why this matters: duplicates are expensive (double bookings, double authorizations). Linking is the safe choice when the order already exists. •

3.3 The form sections, top to bottom •

1. Patient (green check means resolved) | Field | Meaning | |---|---| | Name and gender symbol | The matched patient | | DOB and age | Used for identity matching | | ID (e.g., a short code) | The patient's ID in the system • (likely the MRN or the RIS patient ID) | | Phone, address | Contact details | | **Edit** / × | Correct the match or remove it |

2. Select Insurance (a list of the patient's coverages with checkboxes) | Field | Meaning | |---|---| | Payer name (e.g., a health plan) | The insurance company | | Number on the right | Member / policy ID (or a price for self-pay) | | **"PATIENT PAID IN FULL AT TOS"** with an amount | A **self-pay plan**. • *TOS = Time Of Service*. The patient pays the listed price (here 99.00) when they come in. Self-pay or cash is treated like an insurer in the list. | | Address line | The payer's mailing address | | **Last Checked: N/A** | **Eligibility has never been verified** for this coverage | | **Click to Verify** | Starts the **eligibility check** (270/271 through the clearinghouse, • Availability) | | **Payor Name / Policy Number / Add** | Add another coverage manually | | Checkbox | Choose which coverage attaches to this order |

3. Provider (the ordering doctor)

- **Provider** search by name, or **Search NPI** (the 10-digit ID).
- **New** creates a provider that does not exist yet.

4. Ordered Services

- **Service:** search a **Healthcare Service** from the center's **Compendium** (• the catalogue of bookable exams).

5. Priority

- A toggle, currently **Routine** (the alternative is STAT, meaning urgent).

6. Red warning line. *"Please fill all required fields to create an order: Patient, Provider and Service."* **Provider** and **Service** are shown in red. • This is exactly the rule from the guide: an order is only created when **patient, referring provider and exam** are all resolved. Until then, the Create button stays greyed.

3.4 What this particular fax is • (an important teaching example)

The fax in your screenshot is a **"Request for Release or Transfer of Images and Reports"** from a mammography company. It is **not** an exam order. It asks the imaging center to send a patient's *previous* images to another facility, with the sender's own facility details hand-written in.

What this teaches:

1. **The AI's Category says "Order", but the real meaning is a records-release request.** Categories are a first guess. This is why **Human Review** exists.
 2. **That is why Provider and Service stay empty.** There is no ordering doctor and no new exam on the page. The correct next action is probably not "Create & Complete", but linking it to the patient and routing it to whoever handles **release of information (ROI)**. •
 3. **Why would a facility want prior images?** In mammography a radiologist compares today's images with earlier ones. • That is why a new facility asks the old one to send them.
 4. **Handwriting matters.** The facility name, phone and fax on the form are hand-written. • Roughly 30% of orders arrive handwritten, which breaks conventional OCR. A reviewer should check hand-written digits against the extracted fields (a handwritten 6 and 8 are easy to confuse).
 5. **PowerShare** is mentioned on the form as the electronic image-sharing route. • It is an image-sharing network, outside Gravity's scope. Remember: **Gravity never stores or reads images.**
-

Part 4. The process, step by step (document to order)

Each step lists what happens, who or what does it, what the screen shows and which terms apply.

Step 1. A document arrives

- **Who:** the sender (referring office, patient, staff, another system).
- **How:** one of the seven **sources** (Part 2.5).
- **Screen:** a new row appears in **Documents**, and the **"1 new document"** pill shows.
- **Terms:** *Inbound Fax, Mobile Upload, Web Upload, Document Sync, Source.*
- • **Manual world:** a person watches the fax queue, prints or opens each fax and works out what it is.

Step 2. A document record is created

- **What:** Gravity stores the file (PDF or image) and a record with **Created** time, **Source** and **File Type**.
- **Screen:** columns *Source, File Type, Created*.
- **Terms:** *Document, File type.*

Step 3. The AI classifies it (Category)

- **Who:** Document Center AI. • "Classifies incoming documents."
- **Screen:** the **Category** column fills in (e.g., the blue **Order** pill). Category Status becomes **Auto Assigned**, or **Human Review** if confidence is low. •
- **Terms:** *Category, Category Status, Auto Assigned, Human Review.*

Step 4. The AI extracts the details

- **Who:** AI using computer vision, OCR and an LLM. •
- **What it reads:** patient name, DOB, referring physician, **CPT** (what exam), **ICD-10** (why), payer, subscriber ID. It works on both printed and handwritten text.
- **Screen:** the **Patient Name** and **Birth Date** columns fill in; the review form pre-fills.
- **Terms:** *OCR, LLM, CPT, ICD-10, subscriber ID, structured fields.*
- **Why not just OCR?** • Plain OCR returns raw text and fails on handwriting. Gravity returns *structured fields attached to the correct patient.*

Step 5. Patient linking

- **What:** the extracted name and DOB are matched to an existing patient chart. If none exists, a new patient is created, or a human decides.
- **Screen:** a **chain-link icon** appears next to the patient name. The filter **Patient Linking = No** lists documents not yet matched. In the review screen, the **Patient** card shows a **green check**.
- **Settings:** • **Auto Patient Creation** (on means new patients are created automatically; off means a human decides).
- **Terms:** *Patient linking, patient chart, MRN, stable ID vs fingerprint.* • Matching by name, DOB and phone is "fingerprint" matching and can collide. A stable ID is safer.

Step 6. Insurance is attached and checked

- **What:** the coverage on the document, or already on the patient, is selected. Staff (or automation) can **Click to Verify** to run **eligibility**.
- **Screen:** the **Select Insurance** list with **Last Checked** and **Click to Verify**.
- **Result later:** a green **Active Coverage** badge (seen on the printed order, Part 5).
- **Terms:** *Coverage, Eligibility (VOB), 270/271, clearinghouse, Availity, subscriber ID, COB (primary / secondary), self-pay, TOS.*

Step 7. The ordering provider is resolved

- **What:** the doctor's name or **NPI** is matched to a provider record. If missing, use **New**.
- **Screen:** the **Provider** section (name search and NPI search).
- **Terms:** *Referring / ordering provider, NPI (Type 1 = individual, Type 2 = organization), taxonomy code (specialty).*
- **Why it matters:** referring doctors generate the center's demand, and payers require the NPI on the order and the claim.

Step 8. The exam is matched to the catalogue

- **What:** the requested exam text (or CPT code) is matched to a **Healthcare Service** in the **Compendium**.
- **Screen:** the **Ordered Services** section.
- **Terms:** *Healthcare Service, Compendium, CPT, modifier, protocol.* • Keeping the Compendium free of near-duplicate names is what makes this matching work.

Step 9. Priority is set

- **What:** Routine or STAT.
- **Screen:** the Priority toggle.

Step 10. Duplicate check, then Link or Create

- **What:** Gravity shows **Matching orders found (N)**. A person (or Auto Order Creation) chooses:
 - **Link & Complete** if the order already exists, or
 - **Create & Complete** if it is new.
- **Rule:** • **Auto Order Creation** only fires when patient, provider and exam are all confidently resolved. Otherwise the document goes to a human.
- **Recommended start:** • the **draft order workflow**, where a human confirms Gravity's picks before an order is created. Customers move to Auto Order Creation after trusting the output.
- **Terms:** *Auto Order Creation, draft order workflow, activation (a newly auto-created "shell" record becomes usable once identifiers are confirmed).*

Step 11. The Order exists

- **What:** a record with patient, provider, service, priority, coverage and status.
- **Screen:** the document row now counts as **Order created = Yes**, and the order appears in **Orders**.
- **Why it is "the most important record":** • scheduling, clinical work and every financial step hang off it.

Step 12. Everything after the order

These are covered in the walkthrough. Your printout shows them (Part 5): **Estimate** → **(Prior authorization if required)** → **Booking** → **Check-in** → **Scan** → **Completed** → **Unread (waiting for the radiologist)** → **Report** → **Charges** → **Claim**.

Part 5. The printed order summary (PDF), explained

The PDF is a one-page **order summary / face sheet** with the same data as the order page.

5.1 Header •

Item	Meaning
River City Imaging Centers - New	The facility (tenant)
Phone, fax, NPI	The center's contact details and its organization NPI (Type 2)
QR code and the code beneath it (GV...)	A scannable code that opens the order. • This GV... value is the visit ID (the same format appears in the Documents list Visit column)

5.2 Patient card •

Field	Meaning
ID	Patient ID in the system
Birthdate / age	For identity checks

Field	Meaning
Height / Weight	Used by the technologist (scanner weight and table limits, dose and contrast planning)
Language ("No Language: Add")	Whether an interpreter is needed
Phone / Home / Email / Address	Contact details. The Flow Center and reminders depend on these.
Small icons (copy, link)	Copy the value; open a link
Bug icon at the top of the card	Report a data issue

5.3 Ordering Provider card •

Field	Meaning
Name	The doctor who ordered the exam
NPI	Their individual 10-digit ID (Type 1)
Organization / Location	Their practice (empty here: "No Organization")
Address, Fax, Phone	Where reports and requests are sent. Fax is still the common return path.
Speciality / Credentials	Empty here ("Add"). The taxonomy code classifies specialty.
Tag icon	Label the provider

5.4 Coverage card •

Field	Meaning
"P"	Primary coverage. COB labels are P / S / T (primary, secondary, tertiary).
Payer name	The insurance company
ID	Subscriber / member ID
Active Coverage (green badge)	Eligibility came back active. The 271 answer was positive.
Address	The payer's claims mailing address
Pencil icon	Edit

5.5 Service and Diagnosis card •

Same as the first walkthrough (Part 2 there). New details:

- Study Status: Unread** (earlier screenshot said **Tech Start**).
 - So the sequence so far is **Tech Start** → **(exam finished)** → **Unread**. **Unread means the images exist but a radiologist has not read them yet.**
- Timeline now includes **Completed, 7 minutes ago, by System**.
 - "By System" most likely means an integration message from the scanner or RIS marked the exam done, since nobody clicked it in Gravity.
- Authorized** is still grey, consistent with a screening mammogram that needs no prior auth.

5.6 The Booking tab: "Please contact Support to unlock this service" •

A padlock message. The **Booking** module is **not enabled** for this order or tenant.

- This suggests this customer has not turned on the Booking Center (Sandra), and **"Booked by System"** therefore likely came from the **RIS** through an **SIU** scheduling message instead. Confirm with your team.

5.7 The answer to an earlier open question •

RCI883325 is the **Appointment ID**. The Orders list has a column named exactly that, and the first row shows the same value. • The "RCI" prefix probably identifies the center's RIS.

Part 6. The Orders list, decoded

6.1 Toolbar •

Item	Meaning
All tab	Every order
Search Order	Free-text search
Filters	Same pattern as Documents filters
Appointment Date: Today	Date filter (here showing today's appointments)
+ Create Order	Make an order by hand (without a document)
Refresh, gear, list icon	Reload; column settings; list view. A calendar view is greyed out. •
Displaying 1–10 of 128, 10 / page, 13 pages	128 orders match

6.2 Columns •

Column	Meaning
Patient Name (link icon)	Linked patient
Appointment ID (RCI...)	The RIS appointment identifier
Service	The CPT code
Service Category	Modality group: MG mammography, US ultrasound
Appointment Start Date	Date and time of the slot
Appointment Status	Completed, Booked, Checked In, Canceled (colour-coded: grey outline, blue, green, red). The clock icon beside it is • likely a status history. A red clock on a <i>Booked</i> row may flag something overdue (• unconfirmed).
Priority	Routine or STAT
Insurance	Payer(s) and member ID. Numbers 1 and 2 next to payers mean primary and secondary coverage (COB) .
Tags, Assignees	Labels and owners

6.3 What the four sample rows teach ●●

Row	Service	Status	Teaching point
1	77067, MG	Completed	A normal screening mammogram, the same order as your first screenshot (RCI883325)
2	76700, US	Booked	CPT 76700 is a complete abdominal ultrasound ●. A health plan with a red clock: scheduled but not yet arrived.
3	77067, MG	Checked In	Two payers numbered 1 and 2: a Medicare plan as primary and a second insurer as secondary. That is coordination of benefits (COB) . ●
4	38505, US	Canceled	CPT 38505 is a lymph-node needle biopsy ●. A canceled appointment is a freed slot (● waitlist backfill is the highest-leverage scheduling task). The payer is a Medicare Advantage plan ●, which prior-auths more aggressively (● payer mix).

Part 7. Who does what: human, AI, or the other systems

Stage	Manual center (95%+ today) ●	Sidekick mode ●	Autonomous mode ●
Receive document	Person watches fax queue	Gravity ingests every source	Same
Classify	Person reads and sorts	AI proposes category	AI decides; human only on Human Review
Extract	Person types into the RIS	AI fills the form; person checks	AI fills and accepts
Link patient	Person searches the RIS	AI proposes match; person confirms	Auto when confident
Create order	Person keys the order	Draft order workflow: person approves	Auto Order Creation
Throughput	~10 orders/hour per scheduler	~50/hour	80–95% of volume with no human

Where the other systems sit:

- **RIS:** the system of record for appointments and the worklist. Gravity sits on top and exchanges **HL7** messages (ADT for patients and coverage, ORM for orders, SIU for appointments, ORU for results).
- **PACS:** images and the radiologist's reading. Gravity never touches images.
- **EMR:** the doctor's chart. The source of e-referrals.
- **Clearinghouse (Availity):** carries the eligibility check behind **Click to Verify**.

Part 8. Terms introduced by these screens (A–Z)

Term	Meaning
Active Coverage	Badge showing the eligibility check found the policy active
Appointment ID	The RIS identifier of the booked exam (e.g., RCI...)
Appointment Status	Booked, Checked In, Completed, Canceled

Term	Meaning
Assignee	The staff member responsible for a document or order
Auto Assigned	The AI set the document category itself
Canceled	Appointment cancelled; the slot becomes available
Category / Category Status	The kind of document, and how that was decided or handled
Chain-link icon	The record is linked to a patient
Checked In	The patient arrived
Click to Verify	Starts an eligibility check
Completed	The exam is finished (scan done)
Create & Complete	Create a new order from the document, then close the document
Document Sync	Automatic document import from another system
Face sheet / order summary	A one-page printout of an order's key data
Hide Snoozed	Hide items someone has postponed
Human Review	A person must check because the AI was unsure
Include / Exclude	Filter toggle: show only these / show all but these
Last Checked	When eligibility was last verified ("N/A" means never)
Letter of Protection (LOP)	An attorney's guarantee of payment in injury cases
Link & Complete	Attach the document to an existing order, then close it
Matching orders found	Existing orders that may already fit this document
P / S / T	Primary, secondary, tertiary coverage
PowerShare	An electronic image-sharing network (outside Gravity)
Priority	Routine or STAT
Process	Open a document to work on it
Release of information (ROI)	A request to send a patient's records or images elsewhere
Saved filter / Daily Practice	A stored worklist view with a count
Self-pay / TOS	The patient pays directly at the time of service
Sparkle icon	The AI action or AI marker
Study Status	Where the exam is clinically: Tech Start, then Unread (a radiologist has yet to read it), and so on
Tenant switcher	Choose which customer organization's data to view
Unread	Images done, awaiting the radiologist's report
Visit ID (GV...)	Identifier of the encounter, also printed as the QR caption

For everything else (CPT, ICD-10, eligibility, prior auth, COB, Compendium, Auth Rule, HL7 and so on), see the glossary in Book 1 (Part 6).

Where your team's glossary connects to this stage. The data points from your team glossary (**ordering provider, NPI, payer, subscriber ID, ICD-10, CPT**) are the exact fields the review screen asks for in Steps 4 to 8. If one is empty or wrong at this stage, the later **PASQ** (check for an existing auth) and **APAS** (assisted submission) cannot succeed. Full mapping of each term to its screen: Book 1, Part 9.

Part 9. Talking like an insider about this stage

- "Where do your documents come from, mostly **inbound fax, Document Sync**, or uploads?"
- "What share of documents land in **Human Review** versus **Auto Assigned**?"
- "Are you on the **draft order workflow** or **Auto Order Creation**?"
- "What do you see when **patient linking** is **No**? Is it a new patient or a mismatch?"
- "Is your **Compendium** clean? Near-duplicate exam names break matching."
- "Do you auto-verify **eligibility**, or does staff **Click to Verify**?"
- "How do you handle documents that aren't orders, like **release-of-information** requests?"
- "What does your **Daily Practice** worklist look like, and how many are waiting?"
- "Is **Booking** switched on for this tenant, or does the RIS create bookings through **SIU** messages?"

Reading the document screen aloud:

"This is an inbound fax that the AI categorized as an Order. Patient is linked and the insurance is selected, but eligibility was never checked. Provider and service are empty, so Create & Complete is disabled. Seven possible matching orders exist, so we have to check before creating a duplicate. It's actually a records-release request, so it needs to go to Human Review instead."

Part 10. Open questions for your team (updated)

Answered by this batch:

- ✓ **What is RCI883325?** The **Appointment ID**.
- ✓ **What follows Tech Start?** **Completed**, then **Unread** (waiting for the radiologist).

Still open:

1. What is the **full list** of Category Status, Status and Study Status values? (The dropdowns were cut off.)
2. What exactly do the **sparkle button** and **Process** button do (and which one runs the AI)?
3. Does the **"7 matching orders"** match by patient, date, service, or all three?
4. Is **"Completed by System"** driven by an HL7 message from the RIS or scanner?
5. What does the **red clock** on a Booked order mean?
6. What are the **flask** and **bug** icons?
7. When a records-release fax is categorized as **Order**, what is the correct handling (ROI team, tag, snooze)?
8. Who owns the **Gravity User Portal** source and who uses it?
9. What is the **Visit ID (GV...)** exactly, and how does it differ from the Appointment ID?

Built from your screenshots and the printed order. • items are inferences, not documented facts. The Gravity mechanics are from the guide, whose source material is dated May 2026.